Data 205 Lab 8: Standard Error & Confidence Intervals

Today's Content

Review Midterm Topics

Review Concepts from Lecture 8

New Posit Cloud Topics for Lab 8

Preparing for HW 6

Midterm Q1: Hypothesis

A hypothesis is a testable statement that must:

- State a relationship between two or more variables
- Be falsifiable – it can be tested with data
- Opinions, advice and facts are ***not*** hypotheses

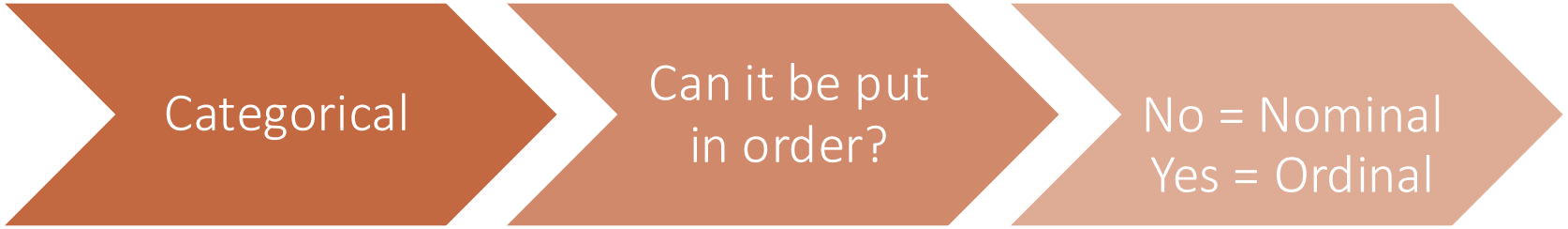


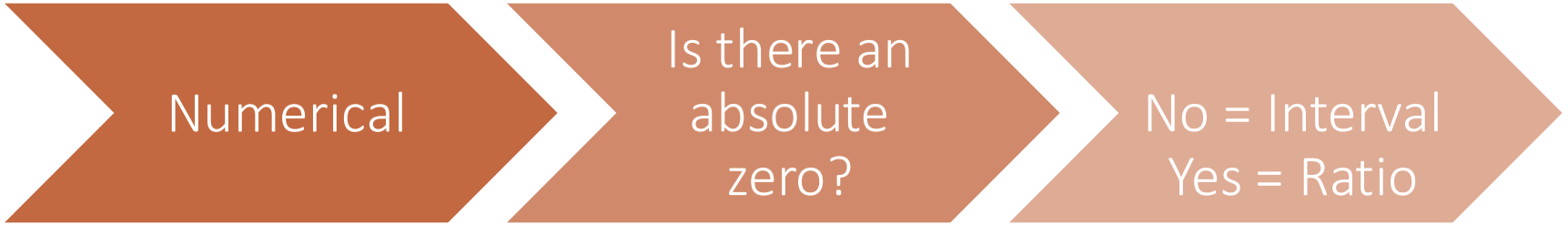
Example:

People who study more tend to have higher exam scores

Midterm Q2: Levels of Measurement

Two questions can tell you the specific level of measurement:





Midterm Q3: How to describe a variable

The specific level of measurement determines what tools you can use:

Level	Central Tendency	Dispersion	Graph
Nominal	Mode only	Percent Distribution	Bar plot
Ordinal	Mode, median	Percent Distribution, Range, Quartiles	Bar plot
Interval/ratio	Mean, median	Range, Quartiles, IQR, SD	Histogram, box plot

Midterm Q4: Representative Probability Sample

Random sample

- Every case has a known chance of selection

Representative:

- Mirrors the population it was drawn from

Not random

- Self-selected and convenience samples

It’d be better if we still go over the midterm questions one by one – we don’t need to show the comprehensive question wording or the R output, but it’d be great to make one slide per question so we can go over them quickly. (students have the PDFs of the midterm anyway)

I revised the first few slides as a reference, and the rest of the midterm review was removed to the bottom of the slide set in case you want to use some of the content.

Review Concepts from Lecture 8

Sampling basics

Population: an entire group of cases that a researcher is interested in

Sample: a subset of the population that a researcher collects data from

Population parameter: characteristics of a population, such as the population mean and population standard deviation.

Sample statistic: characteristics of a sample, such as proportions of a categorical variable, or the sample mean or SD of a numerical variable.

Sample size: the Central Limit Theorem (CLT) requires a sample to be **1) random** and **2) large enough (n>30)** to be able to do statistical inference.

Sampling Error & Standard Error (SE)

- **Sampling error:** the natural variation you get when you take a random sample. An estimate based on a sample **will (almost) always differ from the "true" population value**.
- **Standard error (SE):** The standard deviation of a sampling distribution. It is a way to quantify and measure the sampling error with real-world data.
- Larger SE = Possible sample means are more spread out. Indicates that the sample mean from any random sample is more likely to be far from the true population mean.
- Smaller SE = Possible sample means are closely centered around the true population mean. Indicates that the sample mean from any random sample is more likely to be close to the true population.

SE in a **normal** sampling distribution

Central limit theorem: As sample size increases, the sampling distribution becomes more **normal** and the center of the distribution gets closer to the **true population parameter**, ***assuming a large random probability sample.***

68-95-99% rule: ***assuming a large random probability sample, thus a normal sampling distribution:***

- 68% of all sample means lie within **1 SE** (**z-score=1**) of the true population mean.
- 95% of all sample means lie within **1.96 SE** (**z-score=1.96**) of the true population mean.
- 99% of all sample means lie within **2.58 SE (z-score=2.58)** of the true population mean.

SE Formulas

Mean (numerical variable)

$$SE = s / \sqrt{n}$$

s = sample SD, n = sample size. Use when the population SD is unknown (very common).

Percent (categorical variable)

$$SE = \sqrt{(p(100 - p) / n)}$$

p = sample percent, n = sample size

Confidence Interval

- **Confidence Interval (CI):** an interval built with a sample mean and its standard error (SE), which contains the true population parameter with a 95% or 99% chance.
- E.g., there is a 95% chance that the true population mean lies within 1.96*SE from a specific sample mean.

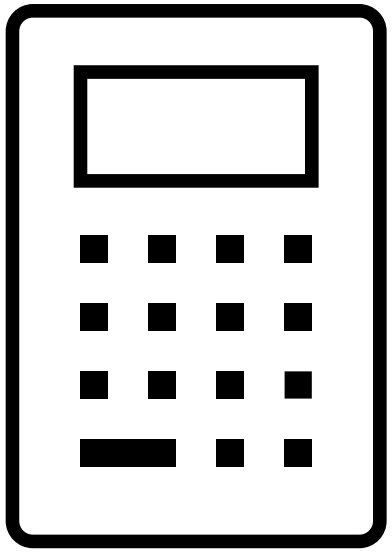
Constructing Confidence Interval

To construct CI:

Step 1: Obtain the sample mean and calculate its standard error (SE).

Step 2: Select a z-score (1.96 or 2.58) based on the confidence level provided (95% or 99%).

Step 3: Plug in the formula to calculate the lower bound and upper bound of the confidence interval (the range of values)



95% CI = [sample mean - (1.96 * *SE*), sample mean + (1.96 * *SE*)]

99% CI = [sample mean - (2.58 * *SE*), sample mean + (2.58 * *SE*)]

Note that confidence intervals are expressed as a range [lower bound, upper bound]

Confidence Levels

Each confidence level has an alpha level (error probability) and a corresponding Z score.

Confidence Level	Alpha	Z Score
95%	.05	1.96
99%	.01	2.58

According to the 68-95-99% rule:
- 1.96 is **the z-score** when CL=95%.
- 2.58 is **the z-score** when CL=99%.

Interpreting Confidence Interval

When interpreting CI, you can use any of the following templates:

Template 1: We are __% confident that the population parameter (mean or percent) falls between __ and __.

Template 2: There is a ___% probability that the population parameter (mean or percent) falls between __ and __.

Template 3: We are __% confident that the estimated range of __ to __ contains the population parameter (mean or percent).

Template 4: There is a ___% probability that the estimated range of ____ to ____ contains the population parameter (mean or percent).

Z-scores or t-scores?

Whether to use Z scores or t scores depends on sample size:

If **n ≥ 100:** Use Z scores: 1.96 for 95% and 2.58 for 99%.

If **n < 100 :** Use t scores from the T table.

Tip: R makes this adjustment automatically!

Degrees of Freedom (df)	Level of Significance for One-tailed Test					
	.10	.05	.025	.01	.005	.0005
	Level of Significance for Two-tailed Test					
	.20	.10	.05	.02	.01	.001
1	3.078	6.314	12.706	31.821	63.657	636.61
2	1.886	2.920	4.303	6.965	9.925	31.59
3	1.638	2.353	3.182	4.541	5.841	12.94
4	1.533	2.132	2.776	3.747	4.604	8.61
5	1.476	2.015	2.571	3.365	4.032	6.85
6	1.440	1.943	2.447	3.143	3.707	5.95
7	1.415	1.895	2.365	2.998	3.499	5.40
8	1.397	1.860	2.306	2.896	3.355	5.04
9	1.383	1.833	2.262	2.821	3.250	4.78
10	1.372	1.812	2.228	2.764	3.169	4.58
11	1.363	1.796	2.201	2.718	3.106	4.43
12	1.356	1.782	2.179	2.681	3.055	4.31
13	1.350	1.771	2.160	2.650	3.012	4.22
14	1.345	1.761	2.145	2.624	2.977	4.14
15	1.341	1.753	2.131	2.602	2.947	4.07
16	1.337	1.746	2.120	2.583	2.921	4.01
17	1.333	1.740	2.110	2.567	2.898	3.96
18	1.330	1.734	2.101	2.552	2.878	3.92
19	1.328	1.729	2.093	2.539	2.861	3.88
20	1.325	1.725	2.086	2.528	2.845	3.85
21	1.323	1.721	2.080	2.518	2.831	3.81
22	1.321	1.717	2.074	2.508	2.819	3.79
23	1.319	1.714	2.069	2.500	2.807	3.76
24	1.318	1.711	2.064	2.492	2.797	3.74
25	1.316	1.708	2.060	2.485	2.787	3.72
26	1.315	1.706	2.056	2.479	2.779	3.70
27	1.314	1.703	2.052	2.473	2.771	3.69
28	1.313	1.701	2.048	2.467	2.763	3.67
29	1.311	1.699	2.045	2.462	2.756	3.65
30	1.310	1.697	2.042	2.457	2.750	3.64
40	1.303	1.684	2.021	2.423	2.704	3.55
60	1.296	1.671	2.000	2.390	2.660	3.46
120	1.289	1.658	1.980	2.358	2.617	3.37
∞	1.282	1.645	1.960	2.326	2.576	3.29

Source: Table III of Fisher & Yates: *Statistical Tables for Biological, Agricultural and Medical Resea* published by Longman Group Ltd., London (1974), 6th edition (previously published by Oliver & Boyd I Edinburgh).

Example 1 (numerical/sample mean)

Example 1: In a sample of 100 students, there is an average exam score of 78 and a standard deviation of 7.5. The population standard deviation is unknown. Using a 95% confidence level, estimate the average exam score of the student population.

Goal: **Estimate the average exam score of the student population (mean)**

$N = 100$ $\overline{X} = 78$ S= 7.5 σ =???

Formula: Since this is a numerical variable, we use $SE = s / \sqrt{n}$

Then, we calculate the lower and upper bounds of CI at 95% confidence level, and interpret.

95% CI = [sample mean - (1.96 * *SE*), sample mean + (1.96 * *SE*)]

Calculation Process

$$SE = s / \sqrt{n} = 7.5 / \sqrt{100} = 7.5 / 10$$

$$SE = .75$$

mean = 78 SE = .75

Since we know confidence interval is 95%:
95% -> alpha .05 -> Z = 1.96. We can then plug this into the step 2 formula.

$$78 \pm (1.96 \times .75) = 78 \pm 1.47$$

Lower: $78 - 1.47 = 76.53$

Upper: $78 + 1.47 = 79.47$

95% CI

76.53 to 79.47

Interpretation:

We are 95% confident that the average exam score of the student population is between **76.53** and **79.47**. There is a 5% probability (0.05 alpha) that it is ***not*** in that range.

Example 2 (categorical/sample percent)

Example 2: In a sample of 100 students, 40% of students received a grade of A or B on the final exam. Using a 95% confidence level, estimate the percent of the student population who received a B or better on the final exam.

Goal: Estimate the percent of student population with a B or better (percentage)

What we know:

(N) = 100 Sample Percent = 40% (95% CI)

Formula: Since this is a categorical variable, we use the below formula to find standard error, then calculate the lower and upper bounds of CI at 95% level, and interpret.

$$SE = \sqrt{\frac{sample\%(100 - sample\%)}{n}}$$

95% CI = [sample percent - (1.96 * *SE*), sample percent + (1.96 * *SE*)]

Calculations

$$SE = \sqrt{(40 \times 60 / 100)} = \sqrt{24}$$

$$SE = 4.90$$

percent = 40 SE = 4.90

95% CI

30.40% to 49.60%

Since we know confidence interval is 95%:
95% -> alpha .05 -> Z = 1.96. We can then plug this into the step 2 formula.

$$40 \pm (1.96 \times 4.90) = 40 \pm 9.60$$

Lower: 30.40 Upper: 49.60

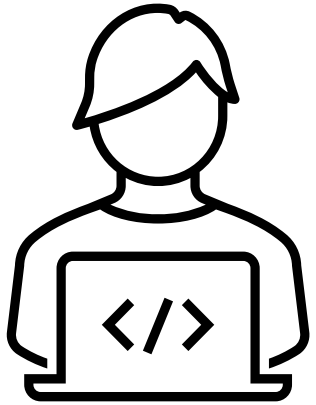
Interpretation:

We are 95% confident that between **30.40%** and **49.60%** of the student population got a B or better. There is a .05 probability that the actual population percent is ***not*** in that range.

New Posit Cloud Topics for Lab 8

Posit Cloud Recap

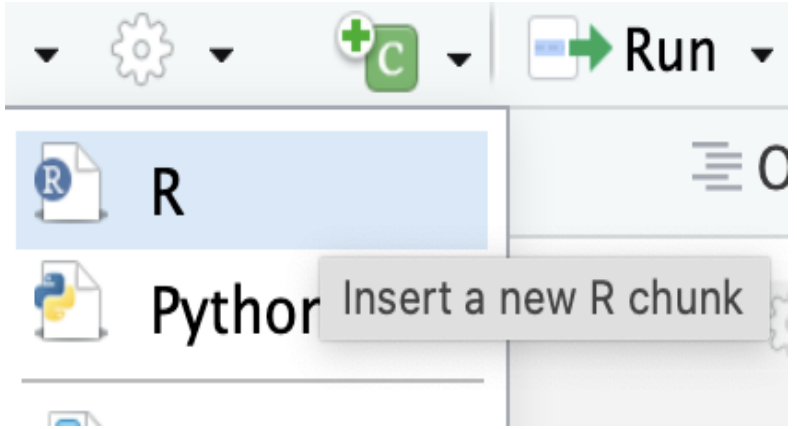
During the last few weeks, we have been introduced to the many technical ways of modifying a markdown file such as running prepared code chunks, modifying the YAML header with your name and current date, and restarting R and running all code chunks if we run into problems. Now it is time to add/edit your own code chunks!

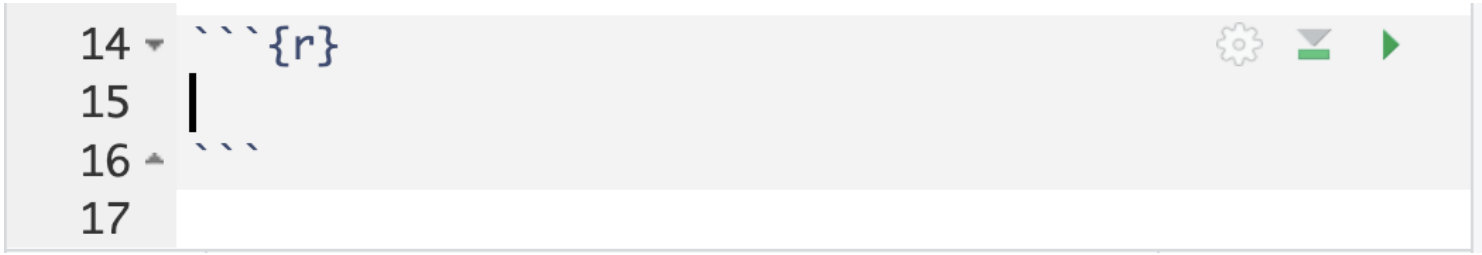


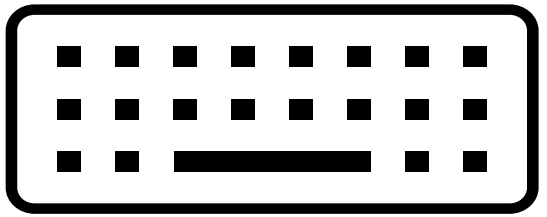
Posit Cloud: Adding a Code Chunk

You can use any one of these three ways:

- Insert button: Click Insert New Code Chunk, then select R
- Type ```{r} and then ``` in the line below it
- Shortcut: Ctrl + Alt + I







Posit Cloud: Deleting, Editing & Commenting

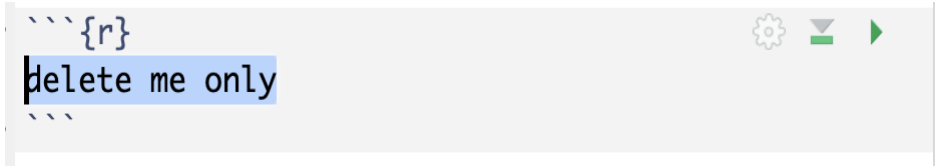
To delete a code chunk

- Highlight all of it and press "delete" on your keyboard. Make sure you highlight all of it including the first and bottom line (```{r}```)



To edit a code chunk:

- You can add or delete any code inside a code chunk but be careful not to edit/delete the top/first line or the bottom last line! (```{r}```) to keep the chunk safe.



To Comment:

- You can add a comment inside a code chunk (or anywhere else in a markdown file) use the # symbol before the comment. Posit Cloud will know to ignore anything to the right of this symbol #.



Posit Cloud: Editing code chunks

```
45
46 ```{r}
47 head(us_state)        |     # return the first 6 rows of the dataset
48 ```
```

Description: df [6 × 12]

	name <chr>	abb <chr>	region <chr>	division <chr>	population <int>	income <int>	illiteracy <dbl>	life_exp <dbl>	murder <dbl>
1	Alabama	AL	South	East South Central	3615	3624	2.1	69.05	15.1
2	Alaska	AK	West	Pacific	365	6315	1.5	69.31	11.3
3	Arizona	AZ	West	Mountain	2212	4530	1.8	70.55	7.8
4	Arkansas	AR	South	West South Central	2110	3378	1.9	70.66	10.1
5	California	CA	West	Pacific	21198	5114	1.1	71.71	10.3
6	Colorado	CO	West	Mountain	2541	4884	0.7	72.06	6.8

6 rows | 1–10 of 12 columns

You can change head to tail if you want to view the last 6 rows of the dataset instead by deleting "head" and replacing it with "tail".

```
45
46 ```{r}
47 tail(us_state) # return the last 6 rows of the dataset
48 ```
49
```

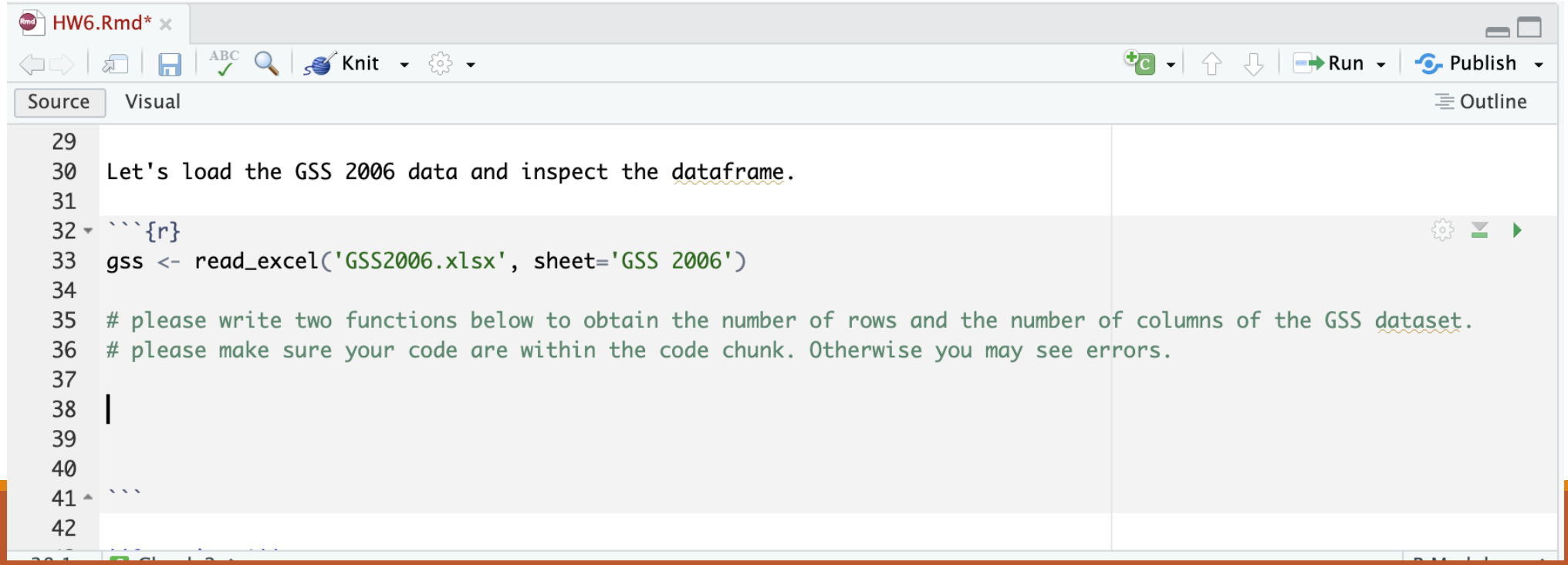
Or you can keep both head and tail, and view by clicking run!

```
45
46 ```{r}
47 head(us_state) # return the first 6 rows of the dataset
48 tail(us_state) # return the last 6 rows of the dataset
49 ```
50
```

Posit Cloud: Write functions

In previous HWs, we've repeatedly used the functions `nrow()` and `ncol()` to obtain the number of rows and the number of columns of a dataset.

Now it's your turn to write the functions and run the code chunk!



Posit Cloud: Reuse functions from previous HWs or tests

Sometimes you can directly borrow a function from previous HWs or notes, and use it on a new variable. You can **copy and paste** the function, then **change the variable name** to the one you need. Then run the code chunk.

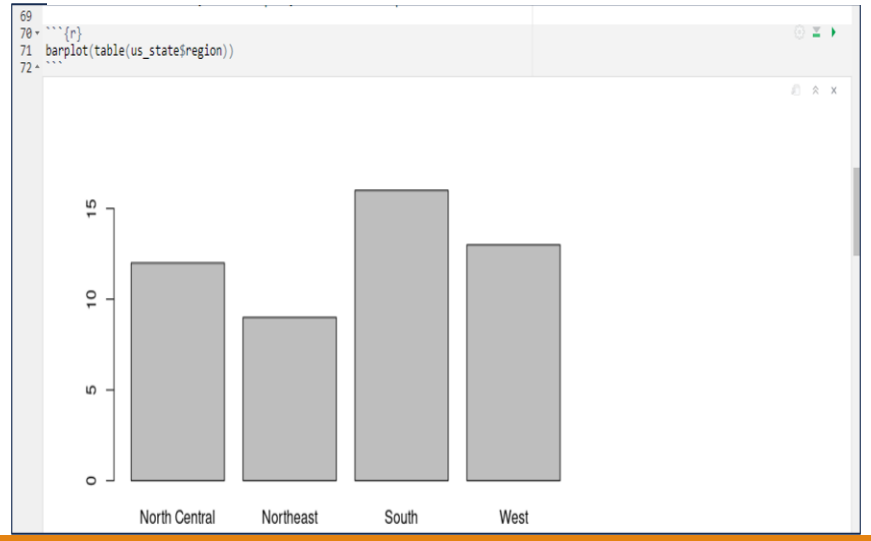
Before

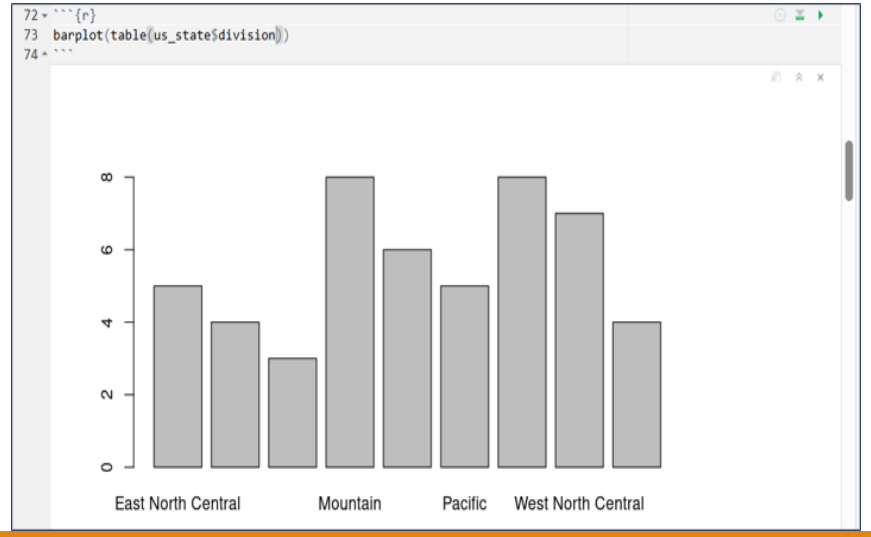
```
barplot(table(us_state$region))
```

→

After

```
barplot(table(us_state$division))
```





Posit Cloud: R as a Calculator

You can use a scientific calculator when constructing confidence intervals and calculating standard error, but R can calculate this as well!

In example 1, we were given N = 100. Sample percent = 40%, with a 95% CI. The goal was to estimate the percent with a B or better. How can we do this in R?

- Type the math in the console.
- Do the calculation step by step.
- Match your hand calculation! The answer should be identical!
- You will **not** be required in this week's homework to insert a code chunk and write out the formulas, but is an optional practice! For your reference, the function to calculate a square root in R is `sqrt()`, and the function to check the number of obs in a variable (sample size) is `length()`.

````
```{r}
7.5 / sqrt(100)
78 - 1.96 * 0.75
78 + 1.96 * 0.75
sqrt(40 * (100 - 40) / 100)
40 - 1.96 * 4.90
40 + 1.96 * 4.90
```
````

Prepare for HW Assignment 6

Discussion for Homework Assignment 6

Go to Brightspace > Course home page > Assignments > HW6 and click this link: (https://posit.cloud/content/7028081)

- Click open the Rmd file from the “Files” pane.
- Follow the steps in the Markdown file
- First code chunk will load three packages
- Second code chunk will read in the GSS file as a data frame. **You need to add some code here.**
- Third code chunk will create summary statistics that will be used to calculate SE and CI of a numerical variable.
- Fourth code chunk will use a new package `plotrix` to replicate the calculation process in R.
- Fifth code chunk will create a prop.table that can be used to calculate SE and CI of a categorical variable
- Finally, change the author and the date, and knit to PDF. Then please go to Brightspace, and upload the PDF.

Discussion for Homework Assignment 6

BrightSpace Assignment “HW6” – **light coding required.**
- 5 points total.
- Deadline: October 25, Sunday, 11:59 pm.
- Submit to BrightSpace. Email submissions not accepted.

Midterm: Data Frames

Data Frame
A table of data containing rows, columns, and cells

Observation (case):
One row; the person, place or thing being measured

Variable:
One column; a characteristic that varies

Midterm: Central Tendency

Mode
The most frequent value. The mode works for any level of measurement

Median
The middle value. Only works when values can be ordered (*Ordinal/numerical)*

Mean
Average of distribution. Numerical only. The mean gets pulled towards outliers

Midterm: Bar Plots .vs. Histograms

Bar plot	Histogram
• Works with categorical Variables • Each bar = a category • Height of bars = frequency	• Numerical Variable • Each bar = an interval of values (width of bar) • Height = number of cases

Midterm: Dispersion

In categorical variables, dispersion is measured by how evenly cases spread across categories

- More concentrated *one* category = low dispersion
- More even = high dispersion

Numerical: You can measure dispersion using range, quartiles, IQR, and SD

Rule

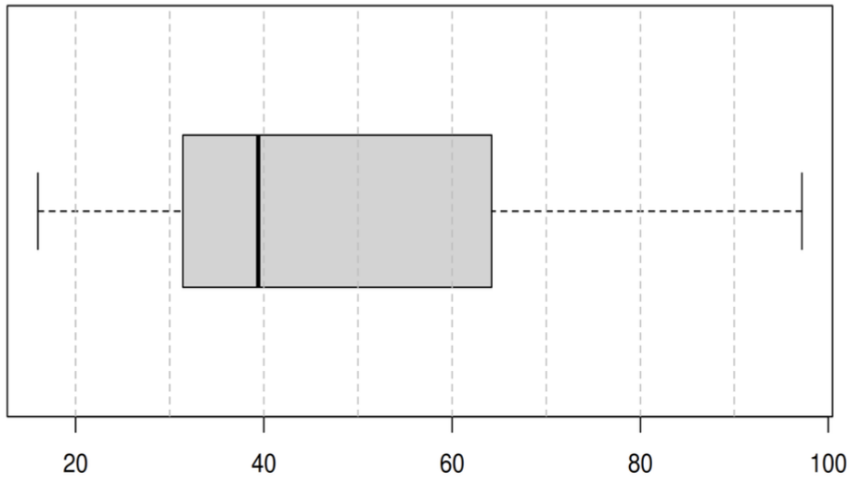
Concentrated = low

Spread out = high

Midterm: Boxplots & Skew

What does a boxplot tell us? **Dispersion**

- Minimum, Q1, median Q3, maximum, IQR = Q3 – Q1 (the middle 50%)
- The median, NOT the mean
- It can be used to compare spread across groups. Bigger box = more spread (IQR)



- Right skew: long tail on the right, when mean > median
- Left skew: long tail on the left, when mean < median

- We discuss skew when looking at distribution curves, box plots, and histograms.

Midterm: SD & Z Scores

Standard Deviation:

- Average distance from the mean. Never negative

Z score

- (X – mean) / SD. Can be positive or negative.

Interpreting Z score

- How many SDs *above* or *below* the mean?